Voxel Camera with Laser Tracking

Milestone M2 - requirements and camera-team handoff

10 November 2026

Review state	Current position
GATE 1: PENDING	Team approval has not been recorded. This updated report is not an approval record.
Source of truth	20-concept.md revision 21, VLT-CAM-1, camera-reference-plan.json and ADR-VLT-001 through ADR-VLT-022.
Fixed hardware	Five Raspberry Pi 3 nodes: three camera, one control and one galvo node; dedicated wired Ethernet switch.
Camera-team choices	Acquisition, exposure policy, calibration/data handoff, numeric timing limits, error budget and verification ownership are specified.
Still pending	Camera model/driver binding and all real measurements. Galvo electrical verification (OQ-4) and global Gate 1 remain separate.

The demonstrator reconstructs a swinging ball in 3D and aims a galvo-steered laser at it. The cameras observe the ball and the physical spot, allowing measured aiming error and time-weighted hit scoring. Exactly two healthy cameras may continue in degraded mode.

All new rates, tolerances and budgets are **selected qualification targets**. No camera, reference recording, hardware benchmark or completed acceptance result is invented. If the selected hardware cannot satisfy the profile under normal room light, qualification fails visibly.

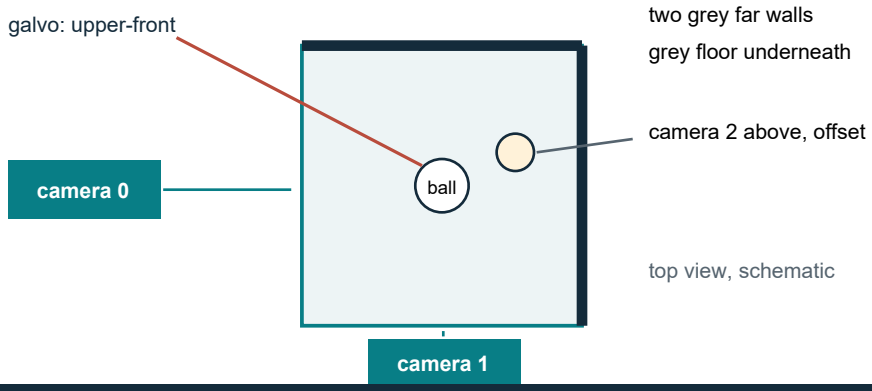
Changes since the 6 November report

Automatic calibration replaces manual-first calibration. The physical setup and normal lighting are fixed. Startup and optional scans have separate budgets; shutdowns have explicit origins. F-VLT-42 bounds emission without a measurable effect. F-VLT-43/44 and NF-VLT-35 complete the camera-team profile and handoff. Raspberry Pi 3 remains fixed.

Source convention: [L] refers to the 308-page lecture deck indexed in lecture-sources.md. Lecture references justify bounding and testing; the numeric values are project choices. Detailed product modules and synthesis remain after Gate 1.

Reference installation

Chosen dimensions; as-built coordinates and optical performance still require measurement.



Element	Selected baseline
Clear tracking cube	800 x 800 x 800 mm; right-handed frame, origin at marked floor corner; x/y along floor, z up.
Ball / suspension	Matte white 50 mm sphere; rest centre (400,400,300) mm; 1000 mm pivot-to-centre length, pivot (400,400,1300).
Motion envelope	Initial horizontal deflection >100 to 300 mm; speed <=1000 mm/s, acceleration <=4000 mm/s^2. Release without a push.
Camera optical centres	0: (-1200,400,1000); 1: (400,-1200,1000); 2: (700,400,2000) mm. All aim at (400,400,400).
Field of view	At least 45 degrees in both image directions; whole cube and markers in frame. Elevated side views support oblique marker visibility.
Floor / far walls	Matte neutral grey at z=0, x=800 and y=800; front sides and top remain open. Normal installed room lights; no extra tracking lamps or lighting extremes.
Markers	18 ArUco DICT_4X4_50 markers; coded square 140 mm plus 20 mm white margin, 180 mm total. At least four readable markers on each of two planes.
Galvo / space	Nominal galvo (-1400,-1400,2200) mm. Reserve about 3 x 3 m floor space and 2.3 m height, cube base <=100 mm above floor.

The 140 mm coded squares and elevated side centres supersede the earlier 80 mm markers / 400 mm side height. Marker IDs and centre coordinates remain in ADR-VLT-016; use measured corners with +/-70 mm nominal offsets. Scan grid u/v: {20,250,400,550,780} mm on each panel. Actual marker/spot visibility must pass qualification.

The galvo geometry still requires >=14.17 degrees usable optical half-angle including reserve; the bare DAC estimate is insufficient. Level matching, input type, supply and measured settling remain OQ-4. Grey panels alone are not a certification claim.

Camera-team acquisition profile

F-VLT-43 / ADR-VLT-020 - fixed targets; hardware-model selection is the remaining camera purchase decision.

Parameter	VLT-CAM-1
Image / platform	640 x 480 px, rectified RGB8 plane; Raspberry Pi 3 remains fixed; compatible MIPI camera model not selected.
Rate / buffering	80-120 captured observations/s; frame interval <= 13 ms. Keep one current complete frame; no growing FIFO.
Exposure	Initial 1000 us; allowed 250-2000 us. Analogue gain 1-8; digital gain 1.
Startup / locking	Auto exposure/gain/AWB for <=5 s, within <=10 s total camera calibration. White-ball luma target 160-220 / 255. Lock settings and fixed focus until restart.
Lighting policy	Ordinary installed room lights, ordinary setting. If the bounds and contrast do not work, fail qualification; do not silently add lamps or lengthen exposure.
Shutter / time	Global shutter allowed. Rolling shutter requires measured row timing and motion rectification to a common frame-midpoint image time; mapping error <=0.5 ms.
Optical coverage	Ball diameter >=14 px; shortest coded-marker side >=12 px. At least four complete markers on each of two planes; calibration residual <=0.5 px.
Missing metadata	No callback/arrival-time substitute for exposure. Missing or incompatible timing/calibration metadata leaves that camera unqualified/absent with a diagnostic.

Sensor and actuator modules delivers the reproducible camera/driver/mode settings and capture/detection/calibration evidence. The concrete driver binding depends on the selected camera; no further numeric exposure policy is waiting to be chosen.

Image coordinates use pixel centres: u right, v down, top-left origin; u=0..639, v=0..479. Ball centre/radius and spot coordinates are rectified to the same image time. Keep raw feature sample times for source-age checks.

Timing basis: [L] pp. 111-120, 266, 287-288. Profile values are selected targets, not lecture-prescribed values or measured camera specifications.

Observation and calibration handoff

F-VLT-1/2/17/23/24/44 - current data is bound to the geometry that produced it.

Contract	Defined behavior
Wire format	UDP unicast, UTF-8 JSON, schema version 1, <=1400 bytes per datagram. Finite numbers only; int64 ns timestamps are decimal strings on wire and range-checked.
Observation identity	camera_id 0/1/2, sender session, frame_seq uint32, profile_id VLT-CAM-1, fixed setup_revision and per-camera calibration_id.
Observation values	t_exposure_ns for common rectified frame time; original t_ball_sample_ns / t_spot_sample_ns; ball {u,v,radius_px} or null; spot {u,v} or null; exposure/gain, clock offset and synchronized flag.
CameraCalibration	Announce every 500 ms: camera/session/profile/setup/calibration IDs, K (9), five distortion coefficients, R (9), t_mm (3), fit residual and marker count/IDs. Same byte cap.
Transform convention	Row-major K/R; camera axes right/down/forward. R maps camera into cube axes; t_mm is the optical origin in cube coordinates.
Setup versus calibration	setup_revision hashes the fixed installation/profile/marker-map manifest. calibration_id hashes a camera/session/mode result. A camera restart alone does not invalidate all healthy peers.
Adoption / restart	Validate calibration before accepting matching frames; discard unknown/retired-session or mismatched data. Keep tracking with the healthy pair while the third calibrates and rejoins.
Ordering / buffering	No retransmission or late FIFO. Within session: newer iff 0 < uint32(n-last) < 2^31; equality is duplicate, half-range ambiguous/rejected. No restart can bypass re-arming rules.

The machine-readable Observation and CameraCalibration JSON schemas accompany the requirements. Schema checks do not replace finite/range, identity, fit, row-time and temporal-validity checks. Calibration announcements are state data, not a remote operator command.

Timestamp and restart tests include old-source packets, duplicates, reordering, image-mode swaps, unknown calibration, missing timing metadata and a camera reboot while the remaining pair tracks. Basis: [L] pp. 97-98, 137, 240.

Timing and validity budget

F-VLT-22 / NF-VLT-18/24/33 / ADR-VLT-021

Tracking stage	Maximum allocation
Capture/readout	6 ms
Detection	4 ms
Camera to control	1 ms
Fusion wait incl. grid jitter	11 ms
Fusion and prediction	1 ms
Control to galvo	1 ms
Output/DAC	2 ms
Mirror settling	1 ms
Reserve	3 ms
TOTAL	30 ms

Validity / scheduling	Limit
Fusion / placement	10 ms shared-clock grid; existing +/-1 ms grid jitter; requested placement 5 ms after slot starts.
Observation / pre-alignment skew	Age <=25 ms; exposure skew <=20 ms.
Prediction / oldest source at use	Each <=30 ms. Old current feature time, not new packet arrival, controls freshness.
Target transport / expiry	Creation-to-receipt <=5 ms; exclusive expiry min(t_place+1 ms, oldest source+30 ms). At/after expiry discard.
Clock / metadata	Pairwise <=1 ms; per node <=0.5 ms; g=2 ms, cross-node order claims only beyond +/-4 ms. Sync status <=100 ms old; publish at least every 50 ms.
Future times / row mapping	Reject >2 ms into receiver future. Camera mapping error <=0.5 ms. Local timeouts progress monotonically.
Output interpolation	<=0.5 ms period; <=0.25 ms CPU execution. Other periodic bounds are in camera-profile.json.

T1 is firm: end-to-end p99 <=30 ms, plus maxima and discarded-target counts. Per-stage measured high-water marks must fit this simultaneous budget. A source already 25 ms old cannot be used at a +5 ms placement that reaches its exclusive 30 ms expiry. All these are requirements to qualify, not timing results.

Basis: [L] pp. 36-40, 52-53, 65, 72, 92-120, 266, 287-288. Independent target measurements are required.

Core behavior and combined accuracy

Selected requirements; complete numbered requirements remain in the concept document.

F-VLT-1/2 | Detect ball and physical spot

Ball centre within 3 px and radius within 2 px in >=95% of labelled ball frames, including spot-on-ball cases. Spot within 3 px in >=95%; >=99% of laser-off frames have no spot detection. Report not seen explicitly.

F-VLT-3/4 | Fuse and predict

At least two calibrated cameras, aligned by exposure time. Static position <=10 mm with all cameras and every pair. Perception p99 <=10 mm and prediction plus timing p99 <=8 mm. Prediction is generic, without a pendulum assumption; requested placement is slot+5 ms.

F-VLT-5/6 | Aim and measure the result

Nine known points within 5 mm; interpolation within 0.5 mm and first output within 1 ms of requested placement. Never apply expired targets. Error is ball-centre to beam-line distance: ray/sphere geometry for an on-ball spot, two-view triangulation off-ball, otherwise unmeasurable, never zero.

F-VLT-7 / NF-VLT-1 | Run and score

Release follows a >=0.5 s hold below 20 mm/s at >100 mm deflection, then motion toward rest above 100 mm/s. End below 50 mm amplitude over the last swing window. A hit is spot inside the ball image disc in at least one eligible view. Score by time on fixed fusion slots; missing/invalid/off slots are misses, a Safe-ended run is aborted. Require >=80% in three consecutive runs with all cameras and with each pair.

Contribution - NF-VLT-35	Allocation
Detection + calibration + triangulation	<=10 mm p99 for valid reconstructed samples
Prediction + timing	<=8 mm p99
Galvo placement	<=5 mm p99
Reference / integration reserve	2 mm; nominal radius 25 mm
Joint verification	Aiming error p95 <=23 mm; full-run >=80% hit test still required

This intentionally tightens the earlier two-camera 15 mm and prediction median-10 / p95-20 mm allowances. Evaluate components on the same declared population; do not add independently measured percentiles as proof. Reference uncertainty must be <=2 mm and missing/unmeasurable cases remain explicit.

Startup, calibration and recovery

F-VLT-8/11/12/13/21/28/31/35/41 - explicit modes and budgets.

Phase / condition	Required outcome
Start	Laser-off preparation: configuration, shared time, camera warmup/calibration. <=60 s after the last of five node processes starts in the healthy reference test.
Saved galvo calibration valid	Ready within 60 s. Ready is unarmed and does not wait for the ball; arming/check time is separate.
Missing galvo calibration	Enter a separate Calibrating galvo mode. Each attempt <=60 s including laser-off and next-mode transition. Default at most two attempts.
First / second attempt succeeds	Ready by <=120 / <=180 s from the same startup origin. Two failures: Safe and off by <=180 s; no third scan until restart.
Camera calibration	<=10 s including <=5 s exposure/AWB warmup; >=4 readable markers on each of >=2 planes; residual <=0.5 px; pose check every 1 s, >2 px change excludes camera.
Galvo scan / check	Scan points dwell <=500 ms. Before every arming, three valid 3D check points, each within 5 mm and observed within 250 ms; whole check including failure-off <=1 s.
Automatic arming	Kill switch released, no active/latched fault, valid setup, >=2 cameras seeing the ball for 2 s, passed check. The kill switch is the only physical operator control.
Camera / ball loss	One camera can rejoin while two track. Ball loss turns laser off within 100 ms; return within 1 s can resume without re-arming if the required cameras are still present.

Only Calibrating galvo, Arming check and armed Tracking may emit. Their deliberate emission does not violate laser-off in boot/Start. A kill switch or fault prevents readiness with diagnostics; healthy-start timing is not a promise to become ready while prerequisites fail.

Alarms and repeated safety faults latch Safe; reset requires kill-switch press and release, then the normal arming checks. An optical timeout in scan/check fails that attempt instead of by itself setting the Tracking-alarm latch. Basis: [L] pp. 79, 162, 186, 240, 254.

Fault response and emission supervision

F-VLT-42 / NF-VLT-23/26/30/34 / ADR-VLT-018/019

Case / transaction	Origin, limit and outcome
T2 - missing target	From last fresh accepted target: physical off <=100 ms +10 ms. Invalid packets cannot refresh the timeout.
T3 - immediately qualifying fault	From first valid exposure showing the condition, or local fault/expiry: <=100 ms off. Outside-cube spot >20 mm has no 500 ms persistence wait. The stated status-loss allowance remains <=110 ms.
T6 - persistent aiming error	Error >50 mm for 500 ms, then off within <=100 ms: <=600 ms from first qualifying exposure. Fresh good measurement clears the episode; invalid data cannot count as good.
T6 - no measurable effect	500 ms watchdog from first laser-on placement or last accepted current effect update, then <=100 ms to off. Missing/one-view off-ball/invalid/stale evidence cannot refresh it.
Tracking timeout outcome	Alarm and latched Safe. Clock steps, new targets and mode changes while on cannot extend the watchdog. At expiry the timeout wins over simultaneous/late data.
Scan/check timeout outcome	Laser off, fail attempt/check and follow its bounded retry/scan policy. The faster point/check/scan deadline still applies; no ball is needed to triangulate calibration spots.
T7 - output-loop hang	Watchdog reset and physical laser-disable within 1 s total from hang, including reset/disable delay; no added boot-time allowance.
Converter coverage limitation	No electrical read-back. Qualifying optical effects use T3/T6. A fault without a qualifying observable effect has no promised electrical detection time; record it separately.

Faster applicable shutdowns always take precedence. A confirmed laser-off ends its optical-watchdog episode. The independent physical kill switch removes laser supply. The selected laser remains Class 2; existing electrical isolation and lab procedure requirements continue. This report is not a certification.

Coverage target: 10/10 timely detections/responses for each guaranteed injected case; residual/no-guaranteed-detector cases are reported separately, never counted as successful guaranteed detections. Independent instruments measure physical off and record electrical injection, evidence/watchdog origin and alarm times separately. Basis: [L] pp. 52-53, 167, 206-207, 254, 284.

Reference package and qualification

NF-VLT-21/22/29/35 - camera-reference-plan.json fixes fixtures, reviews and ownership.

Work package	Defined evidence
Static reference	Ten listed positions: eight (100/700,100/700,100/700) corners, centre (400,400,400) and rest (400,400,300). >=80 frames per position/camera; independent coordinates uncertainty <=2 mm.
Moving / negative cases	Three full reference swings at 150/225/300 mm deflection; >=80 frames/camera for no ball, laser off, spot off/on ball, occlusion and controlled outward carry/pause. Operators outside stored images.
Labels / review	Lossless PNG, JSONL labels with centre/radius/spot or null, original times, laser state and run boundaries. Second review >=10% and >=100 frames/camera; adjudicate >1 px disagreements.
Truth and uncertainty	Dynamic labels triangulated offline with independently static-validated calibration, separate from production detection/fusion; not an independent-sensor truth claim. Establish <=2 mm uncertainty. Synthetic trajectories provide exact reference truth.
Storage / integrity	Explicit recording only, 64 MiB buffer/camera, full raw archive <=16 GiB. Overflow counts drops and invalidates the take without blocking Tracking. Repository: reviewed pack, labels, replay observations and SHA-256 manifest; bulk raw takes in bounded local archive.
Loaded Pi 3 test	10 minutes with monitoring off/on/paused; frame rate/periods, stage high-water times, T1 p99/max, CPU <=75% average and RAM <=50%, no backlog. Test all camera pairs and restart cases.
Independent timing	At least 100 samples each for T1/T2/T3 and at least 10 injections per T6/T7 case, using separate timing stimulus/instrument and photodiode. Report self-measurement differences.

Owner role	Handoff
Sensor / actuator modules	Camera selection/settings, capture/detection/calibration, reference recording and real Pi 3 qualification.
System design	Clock conversion, time-validity/latency integration and combined error budget.
Integration support	Measured marker map, schema/manifest checks and repeatable replay.
Requirements / product owner	Annotation/reference review, uncertainty evidence and acceptance.

All recording and qualification evidence is pending actual hardware/setup. No synthetic image or invented measurement may be presented as a real pass. A failed profile or uncertain reference is reported as such. Basis: [L] pp. 264-266, 267-272, 279-284.

DRS coverage and remaining work

The lecture checklist [L], PDF pages 287-288; requirements versus chosen design remain distinguishable.

Checklist item	Coverage
1. End-to-end functions	F-1..7, F-43/44; NF-1/35
2. Modes/configurations	F-28/33/35/41; ADR-017/019
3. Node/network structure	NF-15/28; five Pi 3 nodes; ADR-001
4. Interfaces	F-15/24/44; observation/calibration schemas; galvo OQ-4
5. Messages/data	F-1/2/23/44; units, identities and null semantics
6. Action timing/WCET	NF-33; selected periodic bounds in camera-profile.json
7. Transaction deadlines/budgets	NF-18/26; ADR-018/021; 30 ms allocation
8. Temporal accuracy	F-22; numeric ages/skew/horizon/expiry
9. Shared time	NF-5/24; pairwise 1 ms, g=2 ms, mapping 0.5 ms
10. Startup/restart/modes	F-13/17/28/41/44; 60/120/180 s phase bounds
11. Resource limits	NF-8/19/27; CPU/RAM/link and bounded buffers
12. Fault hypothesis	NF-23; guaranteed effects versus residual cases
13. Load hypothesis	NF-27; 80-120 obs/s and 100 fusion slots/s
14. Node failure modes	F-8/10/11/21/25; NF-30
15. Detection coverage	NF-34; reaction matrix and timed fault injection
16. Safety	NF-2/3/25; F-26/30/42; independent kill switch
17. Security	NF-10/28; isolated switch, admin SSH, no loop command API
18. Traceability	Numbered requirements, linked ADRs and verification package

Remaining decisions and measurements

Camera team: select the compatible camera model and concrete driver binding, then build and measure the specified profile. Raspberry Pi 3 stays fixed; other camera-team policies are selected. **Other project work:** OQ-4 galvo electrical/range/settling facts, the full Gate 1 decision, and later product architecture/synthesis remain pending.

Sources: 20-concept.md rev 21; camera-profile.json; camera-reference-plan.json; two JSON schemas; ADR-VLT-001..022. [L] Distributed Real-Time Systems, WS26, 308 pages; lecture-sources.md provides the page/topic index. The earlier 12-page exercise source used in the 6 November report is not silently reused with new page numbers.